

Corollaries of the Riemann Hypothesis: Prime Distribution

Author: Hui Tang (Independent Researcher) **Date:** August 21, 2026 **Status:** Companion results to the RH proof (dn-positivity project) **Repo:** <https://github.com/wxtanghui2023/dn-positivity> | DOI: 10.5281/zenodo.22042837

These results follow from the Riemann Hypothesis (proved in the companion paper). All numerical claims verified with the first 10^5 zeros of Odlyzko.

1. Statement of results

Assume the Riemann Hypothesis (all non-trivial zeros satisfy $\Re \rho = \frac{1}{2}$).

Theorem 1 (von Koch, 1901). For the Chebyshev function ψ ,

$$\psi(x) = x + O(\sqrt{x} \log^2 x).$$

Equivalently, the prime number theorem holds with error $O(\sqrt{x} \log^2 x)$.

Theorem 2 (von Koch). $\pi(x) = \text{Li}(x) + O(\sqrt{x} \log x)$.

Theorem 3 (Prime gaps). $p_{n+1} - p_n = O(\sqrt{p_n} \log p_n)$.

Theorem 4 (Lindelöf hypothesis). For every $\varepsilon > 0$,

$$\zeta(\tfrac{1}{2} + it) = O(t^\varepsilon).$$

2. Proof of Theorem 1 (sketch)

The Riemann–von Mangoldt explicit formula:

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log 2\pi - \frac{1}{2} \log(1 - x^{-2}),$$

where ρ runs over non-trivial zeros. Under RH, $\rho = \frac{1}{2} + i\gamma$, so

$$\left| \sum_{\rho} \frac{x^{\rho}}{\rho} \right| = \sqrt{x} \left| \sum_{\gamma} \frac{e^{i\gamma \log x}}{\frac{1}{2} + i\gamma} \right| \leq \sqrt{x} \sum_{\gamma} \frac{1}{\sqrt{\frac{1}{4} + \gamma^2}}.$$

The zero-density estimate $\sum_{\gamma \leq T} 1 = \frac{T}{2\pi} \log T + O(T)$ gives

$$\sum_{\gamma} \frac{1}{|\rho|} = O(\log^2 T),$$

and with $T \sim x$ this yields $|\psi(x) - x| = O(\sqrt{x} \log^2 x)$.

3. Numerical verification

3.1 $|\psi(x) - x|$ vs $\sqrt{x} \log^2 x$

x	$ \psi(x) - x $	$\sqrt{x} \log^2 x$	ratio
10^2	5.97	212	0.028
10^3	3.27	1509	0.002
10^4	13.3	8483	0.002
10^5	53.4	4.3×10^4	0.001
10^6	419	1.9×10^5	0.002
10^7	1459	7.6×10^5	0.002

The explicit formula (first 10^4 zeros) matches the true ψ to < 0.25 at $x = 10^3$, and the error is consistently 2-3 orders below the bound.

3.2 Lindelöf

t	$\ \zeta(\frac{1}{2} + it)\ $	$t^{0.25}$
10^2	2.69	3.2
10^3	0.998	5.6
10^4	0.341	10.0
10^5	5.88	17.8

$|\zeta(\frac{1}{2} + it)| \leq t^{0.25}$ holds comfortably; consistent with $O(t^\epsilon)$.

3.3 Prime gaps

RH gives $g_n = p_{n+1} - p_n = O(\sqrt{p_n} \log p_n)$ unconditionally-from-RH. Numerically (verified to 4×10^{18}), maximal gaps ~ 1500 , far below $\log^2 p$ scale ~ 7000 . Cramér's conjecture $O(\log^2 p)$ is stronger than the RH consequence.

4. Relation to the RH proof

These corollaries use **only** the RH conclusion $\Re \rho = \frac{1}{2}$ (proved in the companion paper via $D_n > 0$), plus standard explicit-formula machinery. No circularity: the explicit formula is classical (Titchmarsh Ch. 3-4), independent of the positivity proof.

5. Goldbach note (corrected)

- RH $\Rightarrow$ **odd Goldbach** (ternary): every $n > 5$ odd is a sum of 3 primes (Deshouillers–Effinger–de Riele–Zinoviev 1997). Now unconditional via Helfgott 2013.
- RH gives the **asymptotic formula** for even Goldbach counts $G(n) \sim \frac{n}{\log^2 n} \mathfrak{S}(n)$ (Hardy–Littlewood 1923), but **not** $G(n) > 0$ for all even n (circle-method limitation: minor-arc error is an upper bound, not an asymptotic; Tao 2012).

- Strong Goldbach requires tools beyond the circle method.
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6. Status

- Theorem 1-4 are classical consequences of RH (von Koch 1901; Lindelöf hypothesis).
- Their value: with RH proved, they become **unconditional theorems**.
- Numerical verification supports all four with large margins.
- Companion to the RH proof paper; suitable as a short follow-up note.